

Reversible Tuning of Visible Wavelength Surface Lattice Resonances in Self-Assembled Hybrid Monolayers

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Periodic arrays of plasmonic nanostructures can support surface lattice resonances emerging from coupling between localized and diffractive modes. This allows the confinement of light at the nanometer scale with significantly increased resonance lifetimes as compared to those of purely localized modes. Here, we demonstrate that self-assembly of plasmonic hybrid nanoparticles allows the simple and fast fabrication of periodic plasmonic monolayers featuring macroscopic dimensions and easily controllable lattice spacings. Electromagnetic coupling between diffractive and localized modes is significantly enhanced when the arrays are embedded in a homogeneous refractive index environment. This is realized through spin-coating of a polymer film on top of the colloidal monolayer. Narrow surface lattice resonances are detected by far-field extinction spectroscopy while optical microscopy reveals a homogeneous coupling strength on cm-sized substrates. The surface lattice resonance position is changed by manipulation of the refractive index of the polymer film through the immersion into different organic solvents. Capitalizing on the thermoresponsive behavior of the polymer film we modulate the surface lattice resonance by temperature in a fully reversible, dynamic manner. The findings demonstrate the potential of colloidal self-assembly as a bottom-up approach for the fabrication of future nanophotonic devices.

Next-generation photonic devices require the concentration of electromagnetic energy to subwavelength scales.^[1] Gold and silver nanoparticles (NPs) could be ideal candidates for this purpose, due to their ability to support localized surface plasmons (LSPs).^[2] Because of large dephasing, mostly due to nonradiative inter- and intraband excitations, LSPs have broad spectral linewidths.^[3] This implies short plasmon lifetimes limiting the extensive use of isolated gold and silver NPs in applications like plasmonic nanolasers^[4] and waveguides.^[5] Nanostructures that support low loss plasmons are in the focus of intense research in recent years.^[6] While addressing the challenge of narrow linewidths on the single particle level requires the optimization of size, shape, and material composition of NPs, a promising concept capitalizing on rather simple building blocks is the arrangement of NPs into periodic arrays. These arrays were found to show narrow surface lattice resonances (SLRs) as result of coupling between LSPs and diffractive modes.^[7] So far, plasmonic arrays

supporting SLRs have been fabricated exclusively using lithographic approaches such as electron-beam lithography^[7b] and soft interference lithography.^[8] However, simple and low-cost fabrication methods suitable for large scale applications that create intense electromagnetic fields with dynamic tunability are yet to be developed.^[9] By exchange of the surrounding solvent some tuning of the SLRs has been demonstrated.^[4b,9] Here, we address a method suitable for fabricating large area devices that support fully reversible, dynamic actuation of SLR modes. We use an array fabrication employing a combination of wet-chemically synthesized hybrid nanoparticles (NPs) and bottom-up self-assembly. cm²-scale hexagonally ordered arrays of silver nanocrystals (NCs) are manufactured. Applying a polymer top layer yields a smart hybrid device supporting pronounced, spectrally narrow SLRs that can be dynamically and fully reversibly modulated. The synergistic interplay between the plasmonic NC array and the responsive polymer layer yields a smart, temperature-controllable optical gate.

Figure 1 illustrates the device preparation based on colloidal self-assembly of silver@gold-poly-*N*-isopropylacrylamide (Ag@Au-PNIPAM) core-shell NPs (a). The NPs have spherical

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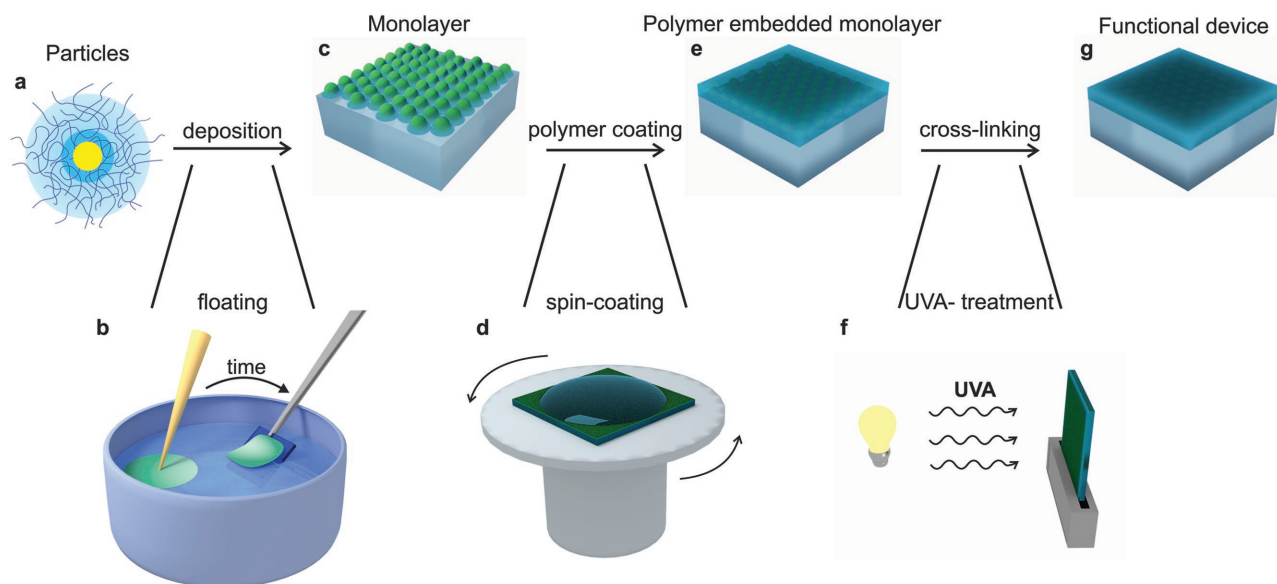


Figure 1. Schematic depiction of the device preparation. a) Ag@Au-PNIPAM NPs with Ag@Au NC cores and PNIPAM hydrogel shells are employed for, b) the colloidal self-assembly at the air/water interface. The freely floating monolayer is transferred to a cm²-sized glass substrate after a certain dwell time. c) The resulting hexagonally ordered monolayer of well-spaced Ag@Au NCs is coated with a PNIPAM layer containing benzophenone by spin-coating (d) resulting in a polymer embedded monolayer (e). f) UVA treatment is used to crosslink the polymer coating forming the final, functional device (g).

gold NC cores of 18 nm in diameter with a silver shell leading to an overall diameter of $d_{\text{core}} = 102 \pm 7$ nm. Due to the large silver shell thickness as compared to the gold size, the Ag@Au NCs behave optically similar to pure Ag NCs.^[10] The optical influence of the Au core can be neglected. These Ag@Au NCs are encapsulated in a semitransparent, nonabsorbing PNIPAM hydrogel shell with a hydrodynamic diameter of $d_h = 326$ nm (at 20 °C). A detailed characterization of the structural and optical properties of the core-shell system is given in the works by Honold et al.^[10] and Dulle et al.^[11] The general NP assembly concept was adopted from our previous work.^[10] Self-assembly of the NPs at the air/water interface (b) followed by subsequent transfer of the monolayer to a hydrophilic glass substrate resulted in a hexagonal array of well-separated Ag@Au NCs (c).^[12] The collected monolayer was dried by applying heat to the substrate in order to firmly adhere the particles to the glass.^[12b] Applying a coating

of linear PNIPAM mixed with benzophenone via spin-coating from 1,4-dioxane (d) leads to an optically homogenous environment nearly matching the refractive index (RI) of the glass substrate, $n_{\text{glass}} = 1.52$ (e).^[7a] A robust device (g) was obtained by photochemically cross-linking the polymer coating with UVA light (f). The resulting polymer matrix is highly cross-linked and strongly bound to the glass substrate.

Figure 2 shows the monolayer characteristics from cm to nm scale. The images a₁–f₁ correspond to an uncovered sample with an interparticle distance $d_{\text{c-c}} = 455 \pm 28$ nm (compare Figure 1c), while the images a₂–f₂ show a similar sample ($d_{\text{c-c}} = 450 \pm 31$ nm) including the polymer coating (Figure 1g). The digital photographs (a) demonstrate the macroscopic dimensions of the prepared samples, here 0.8 cm × 1.2 cm. Optical microscopy images at low (b) and high magnification (c) emphasize the homogeneity as evidenced by the regular color. The

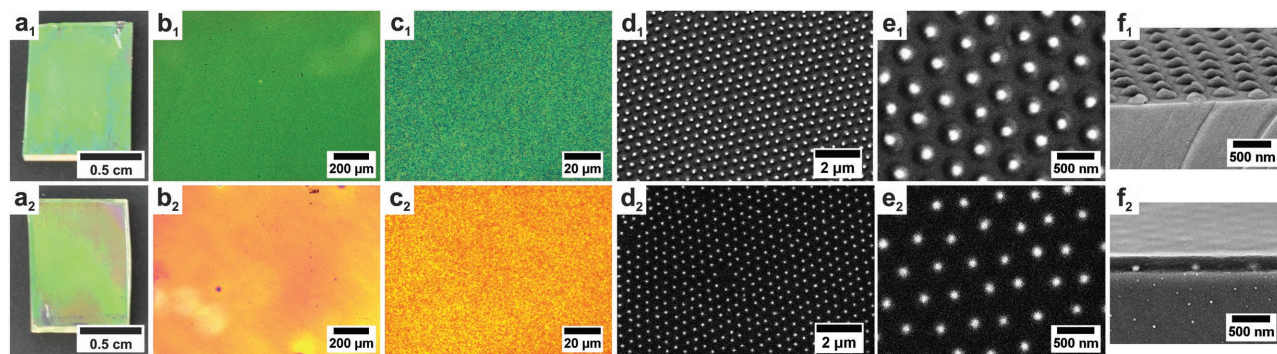


Figure 2. Images of self-assembled Ag@Au arrays investigating cm- to nm-length scales. a₁–f₁) The NC monolayer on glass is compared to, a₂–f₂) the polymer coated device. a) Digital photographs of the samples (0.8 cm × 1.2 cm). b) Optical microscopy images recorded in bright field mode at low and, c), at high magnification. d) Top-view SEM images at low and e) at high magnification. f) Side-view SEM images recorded at a tilt angle of 15°.

granular structure visible in the higher magnification images (c) is related to individual single crystalline domains. It is important to note that the different colors (green and orange) of the uncoated (b_1, c_1) and coated (b_2, c_2) samples are a consequence of the different resonance modes. Figure S1 (Supplementary Information) supports this, showing an optical microscopy image obtained from a polymer coated sample which has been partially treated by an electron beam removing the polymer coating. The polymer coated sample area appears orange while the electron beam treated area appears green. We attribute this change in color to the selective removal of the polymer overcoating. The large single-crystalline domains and the regular hexagonal packing of the Ag@Au NCs are also seen by scanning electron microscopy (SEM) (Figure 2d,e). The regular interparticle distances are a result of the hydrogel shell that acts as a semi-transparent, dielectric spacer between the plasmonic cores.^[10] Comparing the SEM images of the uncoated (Figure 2d₁,e₁) and coated (Figure 2d₂,e₂) samples, it is evident that the monolayer order is not affected by the application of the coating. Figure 2f shows cross-sectional SEM images of the uncoated and coated samples, demonstrating the smoothness of the polymer layer embedding the NC monolayer. Atomic force microscopy (AFM) images obtained from all monolayers relevant to this work along with fast Fourier transforms of the images can be found in Figures S2 and S3 (Supporting Information).

Figure 3 compares experimental (a) extinction spectra from different stages of device preparation to simulated (b) spectra calculated from finite-difference time-domain (FDTD). A dilute aqueous dispersion ($n_{H_2O}=1.33$) of the Ag@Au-PNIPAM NPs (black spectra), which is representative of the uncoupled colloidal building blocks, shows a broad (full width at half maximum, FWHM = 200 nm) LSP resonance at $\lambda_{LSPR} = 505$ nm. Upon assembly of the NPs into an ordered monolayer (Figure 1c), a much narrower (FWHM = 100 nm) and red shifted (to 545 nm) resonance is measured, in good agreement with the FDTD computations (red spectra). The pronounced reduction in linewidth is due to weak dipole–dipole coupling between NCs in the monolayer.^[12b] This coupling remains weak due to the inhomogeneous RI environment with contributions from the glass substrate, the collapsed PNIPAM shells and air (Figure 1c). The average RI surrounding the particle monolayer is relatively low due to the fact that the monolayer is facing the air environment. This limits the LSP strength of the building blocks.

Furthermore the nonisotropic RI surrounding hampers collective plasmon coupling due to scattering losses and diffraction/reflections at the air/glass interface. While the influence of the homogeneous RI on collective coupling events is certainly complex, it appears that interference between scattering by the plasmonic nanoparticles and reflection at the glass layer may play a critical role.^[13] Following an experimental report by Auguie and Barnes,^[7a] we next covered the sample with a polymer layer in order to make the RI environment more homogeneous. Strikingly, the final device shows two narrow and well-separated peaks at 455 and 594 nm (blue spectra). These resonances are a result of strong coupling that is only possible with the homogeneous RI environment (Figure 1e). The peak at 594 nm has a FWHM = 43 nm leading to a Q -factor of $Q = \lambda/\Delta\lambda = 13.8$. This is an increase in Q of more than five times as compared to the LSP resonance of the particles in dilute dispersion. The peak position and linewidth are in remarkably good agreement with the predictions of an early theoretical work which modeled a very similar system of spherical Ag nanoparticles with a diameter of 100 nm arranged in hexagonal arrays in air.^[7f] Using finite-sized single-crystalline domains, peak widths in the range of 20–40 nm ($Q = 10$ –20) were calculated. These quality factors are notably lower than those of devices with larger interparticle distances which support sharp infrared resonances. For example, the resonances of cylinder arrays with peaks at 800–1000 nm can be as narrow as 5–10 nm ($Q = 80$ –200).^[4b,d] In contrast, more closely spaced nanoparticle arrays exhibit broader peaks due to stronger optical losses in the visible wavelength range. Likewise, the FDTD calculations (blue spectrum, Figure 3b) are in good qualitative agreement with the experimental spectra. However, there are deviations in the peak positions and peak strengths. These deviations are mostly related to differences between our experimental monolayer structure (deviation in d_{c-c} , finite domain size, more complex local refractive index environment) and the perfect hexagonal array used for the FDTD calculations. We want to point out that these extinction spectra were measured somewhat differently than in other works reporting on SLRs in lithographically fabricated periodic NP arrays. Here, we probe a large, multidomain sample area of a few mm² by far-field extinction spectroscopy. Therefore, we spectrally average over many single-crystalline domains. This makes it even more surprising that we observe significantly narrowed resonances in our device as compared to the spectra of the uncoupled building

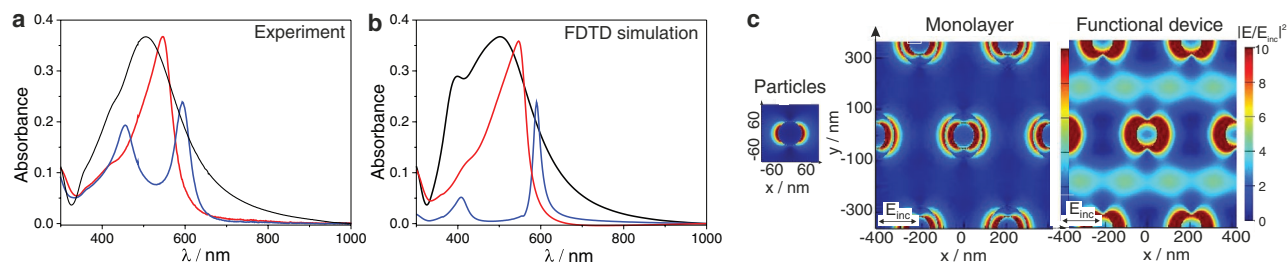


Figure 3. Optical properties of Ag@Au NC arrays. a) Experimental UV–vis spectra show a single LSP resonance at 505 nm for particles in aqueous dispersion (black trace, normalized to maximum absorbance of monolayer on glass), a significantly narrowed and red-shifted LSP resonance at 545 nm for a hexagonal NC array on glass with $d_{c-c} = 419$ nm (red trace), and two distinct peaks at 455 and 594 nm for the NC array covered in a crosslinked PNIPAM top-layer (blue trace). b) Spectra obtained from FDTD calculations with the same color coding as in (a). c) Near-field electromagnetic intensity ($|E/E_{inc}|^2$) maps from FDTD simulation of the particles in water, in an array on glass (monolayer) and in the functional device at the SLR ($\lambda_{SLR} = 594$ nm).

blocks from dilute dispersion (black spectrum, Figure 3a). Insight into the electromagnetic structure of these resonances is provided by the near-field maps from FDTD simulation shown in Figure 3c. For isolated NPs dispersed in water, the near-fields are dominated by the dipolar LSP resonance. In the periodic monolayer, the dipolar fields are still the primary contribution, although they are scattered at the air–glass interface. In contrast, the final device exhibits a pronounced lattice mode at 594 nm with a (1,0) diffraction mode structure, consistent with a SLR. The shorter wavelength resonance peak ($\lambda = 455$ nm) in the spectra of Figure 3a,b points to dipole coupling to higher order lattice modes. In the following we will focus on the investigation of parameters influencing the position and strength of the longer wavelength peak that can be clearly assigned to a SLR. To our knowledge, this is the first report of SLRs in self-assembled, macroscopic arrays of wet-chemically synthesized plasmonic NCs.

The SLR energy can be fine-tuned by immersing a device ($d_{c-c} = 419$ nm) into solvents with a range of RIs, as demonstrated in Figure 4a. The solvents can fully infiltrate the crosslinked PNIPAM coating. The absorbance spectra obtained by a stepwise increase of RI from methanol ($n = 1.33$), ethanol ($n = 1.36$), butanol ($n = 1.40$), chloroform (1.45) to toluene (1.50) show peak shifts from 553 to 600 nm, while maintaining a narrow linewidth (FWHM = 40–50 nm). The observed shift of the SLR closely follows the (1,0) diffraction mode that shifts from 483 to 585 nm as the RI of the environment is increased

from 1.33 to 1.50. For future applications the demonstrated infiltration by solvent can also be used to infiltrate the monolayer with analytes, e.g., for sensing applications. Due to the porous structure of the PNIPAM network the accessibility of the metal cores is still guaranteed.

The soft and deformable character of the PNIPAM hydrogel shell allowed for the fabrication of monolayers with interparticle distances that can be easily tuned by the dwell-time of the monolayer at the air/water interface, producing arrays with spacings from $d_{c-c} \approx d_h$ nm to $d_{c-c} \approx 1.6 d_h$.^[12b] This provides easy access for a systematic study of the coupling strength, i.e., the SLR position and intensity, as a function of interparticle spacing. Figure 4b shows that the SLR peak shifts to lower wavelength for decreasing d_{c-c} and the coupling becomes stronger. For large values of d_{c-c} the LSP resonance at lower wavelengths ($\lambda_{LSP} = 505$ nm) significantly contributes to the spectra, underlining the weaker coupling. The extinction spectra in Figure 4a,b are nicely reproduced with calculations using the coupled dipole approximation (CDA).^[7e,14] The CDA is well-suited to identify SLRs in particle arrays and also allows an efficient screening of array parameters such as the influence of the particle number. Figure S7 in the Supporting Information shows that pronounced SLRs are already expected for hexagonal arrays of only 19 particles. Additional spectra from experiment and CDA simulation for samples with different d_{c-c} as a function of solvent RI can be found in Figure S4 (Supporting

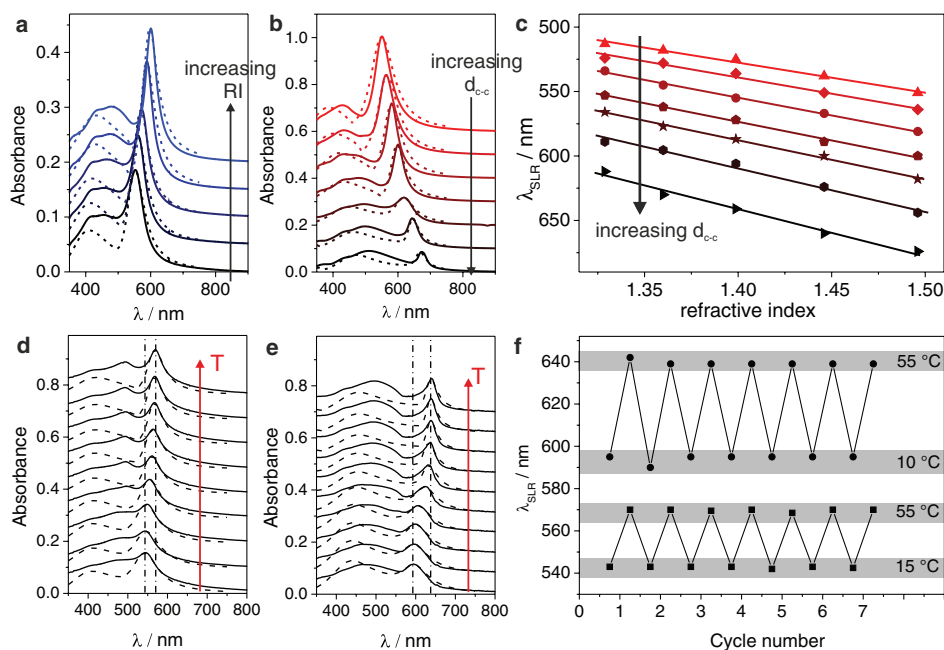


Figure 4. Influence of solvent refractive index, lattice spacing, and temperature on SLR peak position. a) Evolution of the absorbance spectra of a PNIPAM-coated NC array with $d_{c-c} = 419$ nm immersed in different solvents, from bottom to top: methanol ($n = 1.33$), ethanol ($n = 1.36$), butanol ($n = 1.40$), chloroform ($n = 1.45$), and toluene ($n = 1.50$). b) Results from measurements in toluene ($n = 1.50$) in dependence of d_{c-c} , from bottom to top: $d_{c-c} = 506, 474, 419, 407, 374$, and 356 nm. Solid lines are experimental results and dashed lines correspond to the respective results from theoretical CDA computations. The spectra have offsets in steps of 0.1. c) Refractive index dependence of experimentally determined SLR peak positions (symbols). Solid lines are linear fits to data for each lattice spacing (bottom to top: $d_{c-c} = 506, 474, 419, 407, 374$, and 356 nm). d,e) Experimental absorbance spectra for PNIPAM-coated NC arrays measured in water at different temperatures. d) NC array with $d_{c-c} = 430$ nm starting at 15 °C (bottom to top) with temperatures increasing in steps of 5 up to 55 °C. e) NC array with $d_{c-c} = 488$ nm starting at 10 °C (bottom to top) with temperatures increasing in steps of 5 up to 55 °C. Solid lines are experimental data and dashed lines show results from computations using CDA. f) Experimental reversibility study on SLR tuning of the two devices with $d_{c-c} = 430$ nm (squares) and $d_{c-c} = 488$ nm (circles).

Information). Figure 4c combines the results from Figure 4a,b and Figure S4 (Supporting Information). The smallest lattice spacing of $d_{c-c} = 356$ nm exhibits the smallest SLR wavelength shift, from 513 to 551 nm ($\Delta\lambda = 38$ nm). The largest spacing of $d_{c-c} = 506$ nm exhibits greatly red shifted SLR positions, from 612 to 674 nm, and enhanced tuning, $\Delta\lambda = 62$ nm. The tuning behavior of the resonance modes is another indicator for the diffractive/plasmonic coupled nature of the SLRs. The diffraction mode position is proportional to the RI and the lattice spacing ($\lambda \sim n \cdot d_{c-c}$). Consequently, we expect the SLR shift in response to a RI change to be proportional to the lattice spacing ($\Delta\lambda/\Delta n \sim d_{c-c}$). Hence, the same change in RI relates to a more pronounced SLR shift for increasing d_{c-c} , in good agreement with our findings. This study shows that a single batch of core-shell NPs in combination with a simple self-assembly protocol can be employed to produce samples with distinct SLRs in a range of wavelengths as large as 513–674 nm ($\Delta\lambda = 161$ nm). This is possible just by variation of the dwell-time before monolayer transfer, i.e., the lattice spacing, and the RI of the immersing solvent.

In order to create a smart device that can be dynamically tuned with a practically more relevant trigger than solvent exchange, we decided to exploit the temperature responsiveness of PNIPAM. At temperatures above 32–33 °C PNIPAM collapses in water, increasing the polymer density and RI of the top layer. The exact switching temperature depends on the degree of crosslinking and its chemical nature.^[15] This volume phase transition behavior allows us to change the RI environment in an easy, nondestructive manner by simple variation in temperature. Due to the chemical crosslinking of the polymer coating to the core-shell particles, which are firmly attached to the glass substrate, we do not expect the lattice spacing to be affected by the temperature change and the consequent collapse of the polymeric layer. Figure 4d–f shows the influence of temperature on samples with $d_{c-c} = 430$ nm (d) and $d_{c-c} = 488$ nm (e). The sample with smaller d_{c-c} (d) shows a SLR shift from 543 to 570 nm ($\Delta\lambda = 27$ nm), while the larger d_{c-c} sample (e) shows a shift from 595 to 639 nm ($\Delta\lambda = 44$ nm). An example of a temperature cycle in 5 °C increments from 10 to 55 °C for the larger d_{c-c} sample is shown in Figure S5 (Supporting Information). The SLR positions nearly perfectly overlap between the heating and the cooling curve without any indication for a

hysteresis. From the CDA calculations (dashed lines) and comparison to the results shown in Figure 4c, we determine a change in RI from approximately 1.33 to 1.45 through the temperature induced collapse of the PNIPAM layer. It is important to note that this change in RI is mostly determined by the change in water content in the PNIPAM film as water is released from the film upon collapse. Within the investigated temperature window, the RI values of water and PNIPAM individually do not change significantly. The lower RI value (1.33) corresponds to a highly swollen PNIPAM film with an index close to water ($n = 1.33$) at temperatures below the phase transition temperature. The higher RI (1.45) corresponds to a collapsed PNIPAM film with a much higher polymer density and hence an index closer to that of the pure polymer ($n = 1.52$) at 55 °C.^[12b] Evaluating the RI fits from the CDA calculations, the SLR position ($\lambda_{SLR} = 595$ nm) of our device ($d_{c-c} = 488$ nm) under swollen film conditions (10 °C) is quite similar to the static device in methanol with a comparable d_{c-c} ($\lambda_{SLR} = 589$ nm for $d_{c-c} = 474$ nm). This supports the assumption that the highly swollen polymer film leads to roughly the same RI as pure methanol of 1.33, identical to the CDA model fit value. Studying the collapsed state at elevated temperatures, the SLR of 639 nm correlates to an RI of 1.45 when fitting with the CDA method. This corresponds to a polymer film with a water content of 36% (volume fraction). Hence, the PNIPAM film significantly collapses above the phase transition temperature. Under these conditions the sample is optically comparable to the static sample immersed in chloroform. In chloroform an SLR of 624 nm was obtained ($d_{c-c} = 474$ nm). With this comparison a good correlation between temperature increase and RI increase can be qualitatively derived. Figure 4f demonstrates the SLR tuning is fully reversible over 7 heating/cooling cycles without any evidence for depletion, for example, because of device degradation. An extended study on the reversibility of the swelling and collapse behavior with 20 temperature cycles is shown in Figure S6 in the Supporting Information for a comparable sample. Again the reproducibility is very high with no evidence of degradation. To exclude that the SLR position shifts are due to net changes in interparticle distances with swelling of the PNIPAM film, AFM measurements of a device were performed in a liquid cell with water. Figure 5 shows a monolayer on glass in air (a), the same

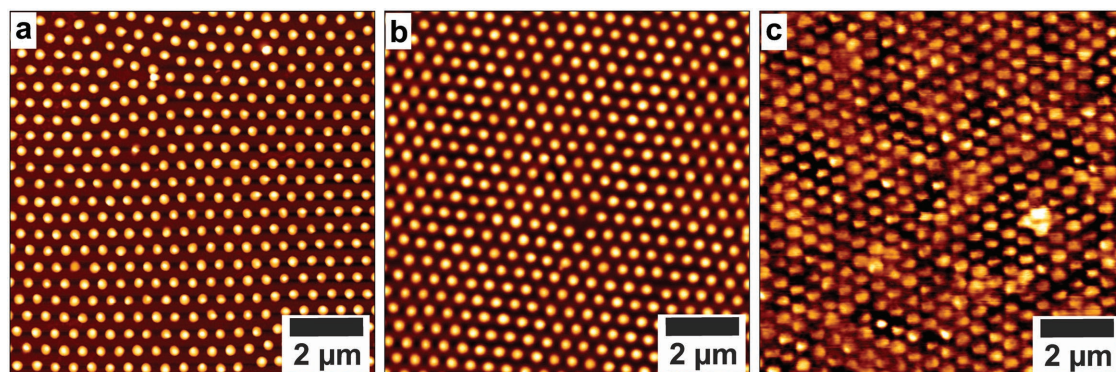


Figure 5. Influence of polymer coating and aqueous environment on interparticle spacing. a) AFM image of an uncoated monolayer with $d_{c-c} = 511 \pm 27$ nm, b) of the same sample with PNIPAM coating against air in the collapsed state with $d_{c-c} = 512 \pm 17$ nm, and c) the swollen state device immersed into water at room temperature with $d_{c-c} = 511 \pm 55$ nm.

monolayer covered with crosslinked and swellable PNIPAM in the dry (collapsed) state (b), and when immersed into water in the swollen state (c). The uncoated monolayer shows an interparticle spacing of $d_{c-c} = 511 \pm 27$ nm. A very similar spacing was obtained for the coated, dry monolayer ($d_{c-c} = 512 \pm 18$ nm) and also for the coated, water-swollen system ($d_{c-c} = 511 \pm 55$ nm) at room temperature. Although the image quality for the measurement in water is not as good as for the dried samples, the particle positions can be clearly identified. These measurements nicely demonstrate that the particle positions remain the same with only the uncertainty increasing from 27 to 55 nm (uncoated monolayer versus device measured in water). The latter change is mostly attributed to the lower image quality for liquid AFM.

In summary, we have provided the first example of SLRs in self-assembled monolayers of wet-chemically synthesized plasmonic NCs. Our lithography-free method is scalable and allows for the facile fabrication of macroscopically coated substrates. The resonance of the SLR strongly depends on the interparticle spacing d_{c-c} and the refractive index of the surrounding environment. In addition, we have demonstrated a new concept of dynamic tuning of the SLR energy in a non-destructive manner in a smart hybrid device. Employing temperature as a reversible trigger to adjust the SLR position, the spectral overlap of the device SLR with other optical modes can be varied. This opens the pathway for dynamic interactions with optically active materials, such as dyes or quantum dots. This new concept may even result in advanced logical photonic components such as switchable nanoscale light sources. Using laser heating of the plasmonic NCs may lead to a local variation of temperature. Consequently, the SLR mode could tune in and out of the absorption resonance leading to a self-regulating optical filter.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

Acknowledgements

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